

Instructor: Michael Sprague, Assistant Professor, School of Natural Sciences
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Lecture Time & Location: MW 2:00 pm – 3:15 pm, COB 261

Office Hours: Monday & Wednesday: 3:15–4:00 pm (S&E 332)

Alternatively, students are welcome to e-mail, call, or just stop by my office. I am rarely on campus on Tuesdays & Thursdays, but can usually be reached via email or cell phone.

Course Webpage: Resources and homework assignments are located at
<http://faculty.ucmerced.edu/msprague/math232.f2008.html>

Prerequisite: Math 231 or Consent of Instructor

Course Units: Four

Catalog Description: Fundamental methods presented in Math 231 are used as a base for discussing modern methods for solving partial-differential equations. Numerical methods include variational, finite element, collocation, spectral, and FFT. Error estimates and implementation issues will be discussed. A significant amount of programming will be required.

Course Goals: We will discuss three broad classes of numerical methods that are very common in science and engineering today for finding approximate solutions to partial-differential equations (PDEs). These methods are (i) finite-difference methods, (ii) spectral methods, and (iii) finite-element methods. These methods will be applied to standard elliptic, parabolic, and hyperbolic PDEs. At the end of this course, students will

1. understand the fundamental theoretical underpinnings of the methods discussed
2. understand the relative strengths and weaknesses of the methods in the context of the three PDE classes
3. understand the importance of validation of numerical methods
4. understand the concepts of consistency, stability, and convergence
5. be able to effectively implement the numerical methods discussed

Learning Outcomes: At the end of this course, students should be able to

- describe typical solution behavior for the three different PDE types (supports goals 2, 5);
- classify a PDE, and choose an appropriate and efficient numerical method to find an approximate solution (supports goals 1–3, 5);
- write a computer program that implements the numerical method (supports goals 1, 3, 5);
- choose an appropriate discretization level for the desired accuracy (supports goals 1, 4, 5);
- demonstrate the validity of an approximate numerical solution (supports goals 3, 5);
- present effectively numerical results in written and graphical form (supports goals 1–3).

Assessment & Scoring: The course grade is based on homework (85%) and class participation (15%). This point distribution was chosen such that, for a student to earn an A, he or she must participate in lectures by asking and answering questions. If a student has overall unsatisfactory class participation, I will email the student asking for more involvement. Please let me know if you have any questions regarding your participation level.

Each homework assignment will be scored on a scale of 100 points. Letter grades will be given in the **approximate** framework: A: 90-100%, B: 80-90%, C: 70-80%, D: 60-70%. Homework will be assigned approximately every two weeks with a total of six or seven assignments. Late homework will not be accepted without prior permission.

Homework Requirements: When preparing your homework, present your answers in a manner that demonstrates your understanding of the concept at hand. To this end, try to understand the point behind me asking you the question or presenting you with the task. If you don't understand the point, please come talk to me and we can discuss it further.

Homework presentation is expected to be of high quality in content, technical writing, and presentation. Please heed the following when preparing homework:

- Homework should be computer generated (i.e., prepare your homework with Word or LaTeX; If you are not already familiar with the use of LaTeX, I highly encourage you to use this course as an opportunity to learn it).
- Present data graphically whenever possible.
- All graphs & figures need to be numbered with appropriate captions and properly annotated (legends, labeled axes, etc).
- Write up conclusions that you draw from any graphs and/or figures.
- All programs should be included as appendices to the homework.
- Please try to layout your homework so that paper use is minimized. For example, don't use a whole sheet of paper to print a three-line code.
- You are encouraged to work in groups. However, **all work turned in must be your own**. You must **list explicitly any outside sources employed** (e.g. websites, *Mathematica*, books, etc.) for each problem you solve. This does not mean that you are allowed to copy a solution should you find it posted elsewhere (see Academic integrity below). All programs must be written by you; no "copying and pasting."

Textbooks: There are no required textbooks for this course. However, reading assignments will be taken from the list of titles below, which are held on a non-circulating basis in the library.

- Pinchover & Rubinstein, *An Introduction to Partial Differential Equations*, Cambridge University Press, 2005
- Strikwerda, *Finite Difference Schemes and Partial Differential Equations*, SIAM, 2004
- Canuto et al., *Spectral Methods: Fundamentals in Single Domains*, Springer, 2006
- Elman, Silvester & Wathen, *Finite elements and fast iterative solvers: with applications in incompressible fluid dynamics*, Oxford University Press, 2005. (The first chapter is available as a free online sample; chapter 1 is all we may cover: <http://www.oup.co.uk/pdf/0-19-852868-X.pdf>)

Other Useful References:

- Boyd, *Chebyshev and Fourier Spectral Methods: Second Revised Edition*, Dover, 2001
- Gottlieb & Orszag, *Numerical Analysis of Spectral Methods: Theory and Applications*, SIAM, 1977
- Iserles, *A First Course in the Numerical Analysis of Differential Equations*, Cambridge University Press, 2000
- Oden & Reddy, *An Introduction to the Mathematical Theory of Finite Elements* (\$16 version to be released by Dover on Aug 28)

Computers, Software & Programs: You may use a calculator (graphing or otherwise) or other computational tool (e.g., Mathematica, Maple, Matlab, etc) to aid in your completion of homework assignments.

Your homework will require a significant amount of programming and data presentation. While any programming language will do, I recommend either Matlab or free alternative to Matlab (Octave or python). However, there is at least one assignment where I will ask you to use the FFT feature in Matlab. When writing your programs,

- use many subroutines;
- write many comments.

Dropping the Course: Please see the UC Merced *General Catalog* for more details.

Special Accommodations: If you qualify for accommodations because of a disability, please submit a letter from Disability Services to the instructor in a timely manner so that your needs may be addressed. Student Affairs determines accommodations based on documented disabilities. I will make every effort to accommodate all students who, because of religious obligations, have conflicts with scheduled exams, assignments, or required attendance. Please speak with me during the first week of classes regarding any potential academic adjustments or accommodations that may arise due to religious beliefs during this term.

Academic Integrity: Academic integrity is the foundation of an academic community and without it none of the educational or research goals of the university can be achieved. All members of the university community are responsible for its academic integrity. Existing policies forbid cheating on examinations, plagiarism and other forms of academic dishonesty. The current policies for UC Merced are described in the Academic Honesty Policy. Go to <http://studentlife.ucmerced.edu/> and look under “Student Judicial Affairs.”